

Project Documentation: Waste Segregation Image Classification Module

1. Executive Summary

This project outlines the development and deployment of an Artificial Intelligence (AI) module specializing in real-time waste segregation based on image processing. The system's primary goal is to classify waste items into four distinct categories with high accuracy and low inference latency, facilitating automated sorting. The module is designed to operate on a host computer, receiving image input from a dedicated camera and outputting classification labels to control the **Smart Dustbin** system, which features **four dedicated compartments** for automated physical sorting.

2. Classification Categories and Scope

The system is trained to distinguish between the following four classes:

Category	Description	Examples
Recycling Waste	Clean, non-organic materials suitable for reprocessing.	Paper, Cardboard, Aluminum Cans, Clear Glass Bottles.
Biodegradable Waste	Organic materials that decompose naturally.	Food Scraps, Fruit Peels, Coffee Grounds, Yard Clippings.
Plastic Waste	Various polymer types requiring dedicated processing streams.	PET Bottles (1), HDPE Containers (2), Plastic Bags, Wrappers.
Medical Waste	Contaminated or biohazardous materials.	Used Masks, Latex Gloves, Syringes (Needle caps), IV Bags.

Out-of-Scope/Edge Cases:

- Heavily soiled or composite materials (e.g., greasy pizza boxes) may require manual intervention or be categorized as the dominant material.
- Classification of specific plastic types (e.g., classifying #1 PET vs. #7 Other) is considered a future enhancement.

3. System Architecture and Data Flow

The architecture is split between the physical acquisition layer and the computational AI inference layer. The output of this system directly controls the mechanical actuators (e.g.,

solenoids, gates) within the **4-Compartment Smart Dustbin**.

3.1. Technical Stack

Component	Technology / Framework	Purpose
AI Language	Python 3.x	Core programming environment.
AI Framework	TensorFlow / Keras (or PyTorch)	Model development, training, and deployment.
Computer Vision	OpenCV	Image preprocessing, resizing, and manipulation.
Data Transmission	PySerial / MQTT / Socket	Communication link between the host computer and the sorting device.
Model	CNN (e.g., MobileNetV2 for efficiency)	Feature extraction and classification.

3.2. Data Flow Diagram

- Image Capture:** A high-resolution camera captures the waste item on the conveyor belt.
- Input Transfer:** The raw image is sent to the Host PC/AI Module.
- Preprocessing:** OpenCV handles image resizing (e.g., to 224×224), normalization, and contrast enhancement.
- Inference:** The CNN model processes the preprocessed image and generates a probability distribution across the four classes.
- Decision:** The category with the highest probability is selected as the final classification label (e.g., "Plastic Waste").
- Control Signal:** The Host PC sends the final label (encoded as a simple string or integer, e.g., 'C3') to the external sorting device via the established data transmission channel. This signal directs the item into the correct compartment of the Smart Dustbin.

4. Data Management and Training

4.1. Dataset Requirements

A high-quality, diverse dataset is crucial for the model's success. The dataset must reflect real-world conditions, including variations in lighting, background, item orientation, size, and minor damage/soiling.

- Minimum Target Size:** $\geq 2,000$ images per category (total $\geq 8,000$ images).

- **Data Augmentation:** Techniques will be applied to increase dataset size and variability:
 - Random rotation, translation, and zoom.
 - Horizontal flipping.
 - Brightness and contrast adjustments.

4.2. Training Methodology

- **Split:** 80% Training, 10% Validation, 10% Testing.
- **Optimizer:** Adam Optimizer (Learning rate to be tuned).
- **Loss Function:** Categorical Cross-Entropy.
- **Early Stopping:** Used to prevent overfitting and ensure the model generalizes well to new, unseen images.

5. Performance Metrics and Success Criteria

The module's success will be measured by its ability to perform accurate and fast classification in a production environment.

Metric	Target Value	Description
Classification Accuracy	$\geq 95\%$ (on test set)	The percentage of correct classifications.
Inference Latency	≤ 50 milliseconds	Time taken for the AI model to output a prediction from image input. Crucial for real-time operation.
F1-Score (Macro)	≥ 0.90	A harmonic mean of precision and recall, providing a balanced measure across all four classes.
Throughput	≥ 15 items per minute	The speed at which the entire system (capture + process + signal) can operate.

6. Deployment and Maintenance

6.1. Deployment Environment

The trained model will be converted into a lightweight format (e.g., TensorFlow Lite or ONNX) to maximize efficiency on the Host PC's processing unit (CPU/GPU). The Host PC will interface with the **Smart Dustbin Control Unit**, which manages the four physical compartments corresponding to the waste categories (C1, C2, C3, C4). The model's low latency allows for rapid decision-making to minimize dwell time before the item is sorted into the correct bin.

6.2. Communication Protocol (Example using PySerial)

The Host PC transmits the result as a simple byte or string through a Serial port.

Example Command Structure:

Category	Compartment	Transmitted Data (String)
Recycling Waste	1	C1\n
Biodegradable Waste	2	C2\n
Plastic Waste	3	C3\n
Medical Waste	4	C4\n

The sorting device monitors the serial port for these commands and triggers the corresponding actuator to direct the waste to the correct compartment.

6.3. Maintenance Plan

- **Continuous Monitoring:** Track misclassification rates during operation.
- **Retraining:** A scheduled process (e.g., quarterly) to retrain the model with newly identified edge cases and misclassified images to prevent model decay.